# Patient-specific design of multilayer vascular grafts reproducing non-linear arterial mechanics: Material bases, thick-walled formulations, and neural networks

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**Abstract**

A vascular graft whose distensibility differs from that of the artery it replaces promotes thrombosis and intimal hyperplasia. Published design methods match a single number—the compliance of an average artery. Here we present a method to design, patient by patient, a multilayer graft that reproduces the complete curve of the artery, including its J shape, evaluated under a strict worst-point target criterion ($e_{\max} \leq 3\%$ of the stress or pressure range). Each graft material is idealized as a fiber that bears no load until it reaches its own stretch (truncated line), and the artery is approximated by a sum of such materials, chosen once for the whole population. The method is first validated on the circumferential stress-circumferential stretch curve, $\sigma_{\theta\theta}(\lambda_\theta)$, which depends neither on pressure nor on wall geometry: with 41 human carotid arteries (Masson et al.) and 22,500 synthetic curves, least-squares fitting followed by minimax rounding to integer laminae meets the target in all 41 subjects with 6 materials; a neural network trained on 50,000 curves reproduces that fit in a single evaluation and generalizes to a constitutive law it never saw (excess error of 2.8 to 4.6%). Without any change, the same tissue-agnostic basis reproduces 327 curves from four other studies (common, external and internal carotid arteries, isolated layers, rabbit, patients with plaque, in vivo measurements): at physiological stresses, all of them meet the target with integer laminae. The method is then applied to the pressure-diameter curve, for which the graft is a thick-walled tube: designed with Laplace's law, no subject meets the target once manufactured, whereas with the thick-walled model 40–41 of 41 do. Design becomes a nonlinear integer problem, solved in three steps: initialization, nonlinear least squares, and minimax integer search against the fitted curve. In the linear case the network does not improve on the exact solution; in the nonlinear case, used as a starting point, it avoids the local minima that, with measured curves, lead to unusable designs (worst error from 468.7% down to 2.3%).

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# 1 Introduction

The failure of small-diameter vascular grafts (thrombosis, intimal hyperplasia) is largely due to mechanical factors: disturbed flow, insufficient strength and compliance mismatch between graft and artery [1]. In the porcine aorta, a graft with a compliance-matching layer preserved pulse pressure, wave velocity and distensibility, which deteriorated when the layer was removed [2]; in rats, a graft whose composition was optimized for the compliance of the host vessel remained matched at four weeks, with less inflammation and more contractile smooth muscle [3]. Matching the mechanics of the artery improves the outcome.

Several studies optimize the design to this end: crosslinking time and thickness of a biopolymer [4], layer proportions and thicknesses of a multilayer graft [5], a nitinol reinforcement [6] or a reinforcing coil [7], and degradable scaffolds that remodel into a neovessel [8, 9]. All of them match one number (compliance or distensibility) of an average artery. The response of an artery, however, is J-shaped: soft at low stretch, when elastin and smooth muscle carry the load, and stiff at high stretch, when collagen fibers are recruited [10, 11]. A single number does not describe that shape, and the shape changes from patient to patient and from artery to artery.

This work designs the graft so that it reproduces the complete curve of each artery. The method operates on a population-wide, material-level basis that remains tissue-agnostic across J-shaped arterial responses, bringing in subject-specific geometry and pressure only during tube-level design. The method has two levels, presented in this order:

1. **Material level (validation).** The target curve is the circumferential stress as a function of circumferential stretch, $\sigma_{\theta\theta}(\lambda_\theta)$. It is a property of the tissue: it does not depend on pressure, radius or wall thickness, and is therefore common to any artery whose wall has a J-shaped response. At this level the graft is a sum of materials, the design is linear, and every step of the method can be compared with an exact solution.

2. **Tube level (application).** The target curve is the pressure borne by the wall as a function of diameter. Geometry and the thick-wall model come into play, the design is no longer linear, and the conclusions of the first level are used.

The contributions are: (i) a basis of fiber-like materials chosen once for the whole population; (ii) the exact transition from real-valued amounts to integer laminae by integer linear programming (minimax); (iii) a neural network that yields the design in one evaluation and generalizes to unseen laws; (iv) the demonstration that, at the tube level, the design must be carried out with a thick-walled model, which makes it a nonlinear integer problem; and (v) a three-step procedure in which the network acts as a starting point.

# 2 Data

## 2.1 Constitutive law and subjects

Masson et al. [12] measured non-invasively, over the cardiac cycle, the diameter and pressure of the common carotid artery of 16 normotensive (NT) and 25 treated hypertensive (HT) subjects, and fitted to each one a model of the intima-media wall with three contributions: neo-Hookean elastin, four families of collagen fibers with exponential response (axial, circumferential and two diagonal at $\alpha$) [10], and smooth-muscle tone [13]. With fixed axial stretch $\lambda_z$ and incompressibility,

$$\sigma_{\theta\theta} - \sigma_{rr} = c(\lambda_\theta^2 - \lambda_r^2) + \lambda_\theta^2 \sum_{k=1}^{4} c_{1,k} E_k e^{c_{2,k} E_k^2} \sin^2 \varphi_k + T_m \lambda_\theta \left[ 1 - \left( \frac{\lambda_m - \lambda_\theta}{\lambda_m - \lambda_0} \right)^2 \right], \qquad (1)$$

with $E_k = \lambda_k^2 - 1$, $\lambda_r = 1/(\lambda_\theta \lambda_z)$ and $\lambda_k$ the stretch of each family ($\lambda_z$, $\lambda_\theta$ and $\sqrt{\lambda_\theta^2 \sin^2\alpha + \lambda_z^2 \cos^2\alpha}$). The 14 parameters of each subject are listed in their Tables 2 and 4 (Appendix A). Ten subjects (4 NT and 6 HT) are held out for testing and play no part in any choice made by the method.

### 2.2 Stress curves

The target curve of the first level is $\sigma_{\theta\theta}(\lambda_\theta)$ at the media-adventitia boundary, at 999 values of $\lambda_\theta$ between 1.02 and 1.16, computed with (1); at that boundary the radial stress equals the perivascular pressure of Masson's model. Recomputed in this way, it reproduces the published mean curves (Figs. 5a and 7a of [12]) with a root-mean-square error of 1.2 to 1.7% of their mean value, of the order of the digitization error. The 41 curves are J-shaped (monotonic and convex) and range from 4.7 to 49 kPa at $\lambda_\theta = 1.02$ and from 54 to 290 kPa at 1.16. Since pressure is also expressed relative to 13.332 kPa (100 mmHg), stress is additionally shown relative to a reference stress of 150 kPa. In dimensionless form, $(\sigma - \sigma_{\min})/\Delta\sigma$, normotensive and hypertensive subjects have almost the same shape; the difference lies in the stress level.

Two bounding curves enclose the 41 subjects: $y_1$, a curve of Masson's law with parameters within the range of the subjects, which starts at the lowest measured stress and lies below all of them, and $y_2$, the stiffest subject extended up to 300 kPa. The working space consists of 22500 synthetic curves of Masson's law within that band, with parameters of 2 to 4 training subjects mixed and perturbed, or drawn between the minimum and maximum of the subjects.

## 3 Validation of the method on the stress-stretch curve

### 3.1 Materials: truncated lines

One lamina of material $j$ contributes $A_j(\lambda) = S_j \max(0, \lambda - k_j)$: nothing up to its knot $k_j$, and then a straight line of slope $S_j$. It idealizes a crimped fiber that bears no load until it is straightened. The graft is the sum $\sigma_{\text{gr}} = \sum_j c_j A_j = \mathbf{A}\mathbf{c}$ where $c_j$ is the number of laminae of each material and $\mathbf{A} \in \mathbb{R}^{999 \times N}$ is the basis. For fixed knots, the problem is linear.

**Choice of the knots.** The knots are chosen once, for the whole population, by minimizing a function $J(k)$ that combines three requirements on the working space and the training subjects:

- **Accuracy: CVaR 99%.** For each curve, the maximum pointwise error of the least-squares fit, $e_{\max}$ (Eq. 3), is computed. The CVaR 99% (conditional value at risk [14]) is the mean of $e_{\max}$ over the 1% of curves with the largest error. A smaller term, 0.25 times the mean NRMSE, is added so that the typical fit also improves.
- **Stability: condition number.** With the columns of $\mathbf{A}$ scaled to a maximum of 1, the condition number is $\kappa(\mathbf{A}) = s_{\max}/s_{\min}$ [15]. We require $\kappa(\mathbf{A}) < 30$; the adopted bases have 28.0 (6 materials) and 19.0 (5).
- **Sign.** The amounts of the unconstrained fit, $\mathbf{c} = \mathbf{A}^{+} y$, must be non-negative (with a tolerance of 0.01 lamina) for the 22500 curves of the working space, the training subjects and the bounding curves.

The search is global, by differential evolution [16]. The best candidate is refined with the Nelder-Mead simplex method [17].

### 3.2 Metric

For a target curve $y$ with range $\Delta = \max y - \min y$,

$$e_{\max} = \frac{100}{\Delta} \max_i |y_i - \hat{y}_i| \quad [\%], \qquad E_{\text{rms}} = \frac{100}{\Delta} \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \quad [\%], \tag{2}$$

Table 1: Adopted bases: knots $k$ and slope of one lamina $S$ [kPa].

| | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|
| 6 materials, $k_j$ | 0.8193 | 1.0227 | 1.0774 | 1.1099 | 1.1318 | 1.1478 |
| $S_j$ | 12.32 | 41.39 | 34.73 | 57.79 | 83.23 | 111.13 |
| 5 materials, $k_j$ | 0.8228 | 1.0243 | 1.0881 | 1.1226 | 1.1447 | – |
| $S_j$ | 12.56 | 43.57 | 51.92 | 88.74 | 129.08 | – |

where $\hat{y}$ is the graft curve. Target: $e_{\max} \leq 3\%$.

### 3.3 Real-valued amounts: least squares

With a fixed basis, the best quadratic fit is $c = \mathbf{A}^{+}y$, where $\mathbf{A}^{+}$ is the Moore-Penrose pseudoinverse [18, 19]. The fit meets the target in all 41 subjects, with a worst error of 0.88% (6 materials; median 0.67%) and 1.41% (5 materials; median 1.07%).

### 3.4 Integer laminae: three ways of obtaining them

A graft is manufactured with an integer number of laminae, $z \in \mathbb{Z}^{N}_{\geq 0}$.

- **Simple:** each amount given by the network to the nearest integer.
- **Joint (lowest $E_{\mathbf{rms}}$):** lattice search with Schnorr-Euchner pruning [20].
- **Integer minimax design (lowest $e_{\max}$):** directly seeks the integers that leave the narrowest tube around the curve of the artery via mixed-integer linear programming solved by branch and bound with HiGHS [21].

Table 2: 41 subjects, integer laminae: median / worst $e_{\max}$ [%] and subjects meeting the target.

| | **6 materials** | | **5 materials** | |
|---|---|---|---|---|
| **Method** | **41 subjects** | **2000 curves** | **41 subjects** | **2000 curves** |
| Real-valued (least squares) | 0.67 / 0.88 (41/41) | – | 1.07 / 1.41 (41/41) | – |
| Simple rounding (from network) | 2.34 / 7.45 (25/41) | 1706 | 2.23 / 7.48 (26/41) | 1616 |
| Joint from network | 1.17 / 2.46 (41/41) | 1998 | 1.70 / 3.56 (40/41) | 1997 |
| Joint from LS (Schnorr-Euchner) | 1.18 / 2.46 (41/41) | 1998 | 1.70 / 3.56 (40/41) | 1997 |
| Integer minimax (MILP) | 1.02 / 2.35 (41/41) | 2000 | 1.47 / 3.50 (40/41) | 2000 |

### 3.5 The neural network

A multilayer perceptron [22] $999 \to 256 \to 128 \to 64 \to N$ with ReLU [23], layer normalization [24] and Adam optimizer [25] learns the map $y \mapsto \mathbf{A}^{+}y$. Trained on 50000 curves in seven families. On a law never seen (Fung-type exponential [26]), the network reproduces 300 curves with an excess of +4.6% (6 materials) and +2.8% (5) over the optimum.

### 3.6 External validation: measurements by other authors

Applied to 327 curves from four independent studies: Sommer and Holzapfel [27], Holzapfel et al. [10], Liu et al. [28], and Kamenskiy et al. [29]. At physiological stresses, all accepted curves meet the target with integer laminae.

# 4 Application: the pressure-diameter curve and the thick-walled tube

## 4.1 From stress to pressure

Radial equilibrium gives the pressure borne by the wall [11, 12]:

$$P = P_i - P_a = \int_{r_i}^{r_m} \frac{\sigma_{\theta\theta} - \sigma_{rr}}{r} dr. \tag{3}$$

## 4.2 Thin wall and thick wall

Laplace's law assumes a thin wall, whereas the real carotid artery is thick ($h/r \approx 0.2$). Inner fibers stretch more than outer ones:

$$r^2 - r_i^2 = R^2 - R_i^2, \qquad \lambda(r) = \frac{r}{R}. \tag{4}$$

Pressure in the thick-walled tube is:

$$P = \sum_{j=1}^{N} \int_{\text{layer } j} \frac{\sigma_j(\lambda(r))}{r} dr. \tag{5}$$

Designed with Laplace's law, no subject meets the target once manufactured (worst error up to 32.5%); with the thick-walled model, 40-41 of 41 do.

## 4.3 The thick-walled graft and design methods

The design parameters are $\theta = (h_1, ..., h_N, \delta)$, where $h_j$ is the thickness of layer $j$ and $\delta$ is a common prestretch ($|\delta| \leq 0.25$). The three-step procedure is: (1) initialization via neural network; (2) nonlinear least squares fitting (TRF [30]); (3) integer minimax search against the fitted curve.

# 5 Discussion

Working on the stress-stretch curve first allows validating each component against exact linear solutions. The transition to the tube adds geometry and thick-wall kinematics. In the thick-wall case, the neural network acts as amortized optimization [31], initializing the non-linear optimizer to avoid local minima under noise.

From a clinical and manufacturing perspective, performing the integer search against the smoothed, fitted curve (Step 2) rather than raw measured data establishes an essential protocol when handling sparse or noisy in vivo measurements. Furthermore, the fitted pre-stretch parameter $\delta$ physically corresponds to controlling fiber crimping during fabrication. Modern micro-fabrication techniques, such as multi-layer electrospinning, micro-knitting, or precision dip-coating, capable of depositing $\sim 25\,\mu$m laminae, provide a viable pathway for manufacturing these designs.

# 6 Conclusions

1. A basis chosen once reproduces the stress-stretch curve of 41 carotid arteries and 22500 synthetic curves within 3% worst error with integer laminae.

2. At the tube level, the design must use a thick-walled model: Laplace designs fail upon manufacturing.

3. The three-step procedure initializes via neural network, fits by non-linear least squares, and searches integer laminae against the smoothed curve, meeting the target in 40-41 of 41 subjects in under 0.5 s.

## A Parameters of the 41 Subjects

Tabulated values from Masson et al. [12].

## B Worked Example of a Thick-Wall Design

Numerical illustration showing incompressibility, slice-wise integration, layer positioning effects, and knot shift through prestretch $\delta$.

## C Training of the Neural Networks

Details on learning curves, loss terms, and condition numbers of the inverse problem.

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